Part 1. Exploratory Data Analysis

Group 3, Presentation and Visualization, Infovis2627

The data

The dataset contains 119,390 real bookings of two Portuguese hotels, a resort in the Algarve and a city hotel in Lisbon, with arrivals from July 2015 to August 2017. Each of its 32 fields was classified by meaning (Figures 1 to 4): 15 are categorical (13 nominal, 2 ordinal: month and meal) and 17 numerical (12 discrete ratio counts, 4 discrete interval date parts and the continuous ratio price per night, `adr`). Nine fields are temporal and two geographic.

Quality

Missing values are written as text. `NULL` in `agent` and `company` means "not applicable", while 488 countries are real gaps. 31,994 rows (26.8 %) repeat an earlier row exactly, and without a booking identifier they cannot be told apart from identical bookings. A few values are impossible (180 bookings without guests, a negative price, a price of 5,400) and `reservation_status` repeats `is_canceled` (Figure 5).

Distributions

All numeric fields are right-skewed, the median lead time is 69 days but the mean is 104 (Figure 6), so medians describe them better. The resort's prices vary much more (Figure 7), stays peak at 7 and 14 nights (Figure 8), most categorical fields are dominated by one value (Figure 9), Portugal provides 40.7 % of bookings (Figure 10) and many details are rare (Figure 11). The charts follow the course's charts by data type: histograms and box plots for distributions, sorted bars for categories, and small multiples on a common scale (Gestalt similarity) to compare groups.

Relationships

Bookings and prices follow the season (Figures 12 and 13): the resort's median price rises from 46.8 in November to 180.4 in August. Cancellations (37.0 %) grow with lead time, from 9.6 % to 67.7 %, and are highest for groups (Figure 14). The 99.4 % of non-refundable deposits is probably a recording effect of duplicated blocks. The market mix is similar in both hotels (Figure 15) and all correlations are weak (Figure 16), so they are not causes.

Currency and decisions

The data is a closed historical snapshot, about nine years old, with only 2016 complete; it needs no updates, and a live version would need daily or monthly updates depending on the audience. After the approved cleaning rules, 119,201 rows remain. The dashboard will use hotel, arrival date, cancellation, lead time, price, market segment, country, nights, guests, customer and deposit type, highlighting season, cancellations and origin.

Figures

Field	Meaning	Data type (course taxonomy)	Order	Temporal or geographic
hotel	Hotel of the booking: Resort Hotel (Algarve) or City Hotel (Lisbon). Added when the two datasets of the paper were joined	Categorical, nominal	No order	Geographic (hotel location)
is_canceled	The booking was cancelled (1) or not (0)	Categorical, nominal (binary)	No order	Neither
lead_time	Days between the booking entering the hotel system and the arrival date	Numerical, discrete, ratio	Sequential	Temporal (duration)
arrival_date_year	Year of the arrival date	Numerical, discrete, interval	Sequential	Temporal
arrival_date_month	Month of the arrival date (written as the month name)	Categorical, ordinal	Cyclic	Temporal
arrival_date_week_number	Week number of the arrival date in its year	Numerical, discrete, interval	Cyclic	Temporal
arrival_date_day_of_month	Day of the month of the arrival date	Numerical, discrete, interval	Cyclic	Temporal
stays_in_weekend_nights	Weekend nights (Saturday or Sunday) stayed or booked to stay	Numerical, discrete, ratio	Sequential	Temporal (duration)
stays_in_week_nights	Week nights (Monday to Friday) stayed or booked to stay	Numerical, discrete, ratio	Sequential	Temporal (duration)
adults	Number of adults	Numerical, discrete, ratio	Sequential	Neither
children	Number of children (paying and non-paying)	Numerical, discrete, ratio	Sequential	Neither
babies	Number of babies	Numerical, discrete, ratio	Sequential	Neither
meal	Meal package: SC or Undefined = no meal, BB = bed and breakfast, HB = half board, FB = full board	Categorical, ordinal (meals included)	Sequential	Neither
country	Country of origin of the guest, as an ISO three-letter country code	Categorical, nominal	No order	Geographic (guest origin)
market_segment	Market segment of the booking (TA = travel agents, TO = tour operators)	Categorical, nominal	No order	Neither
distribution_channel	Channel used to make the booking (TA = travel agents, TO = tour operators, GDS = global distribution system)	Categorical, nominal	No order	Neither

Figure 1. Data dictionary A (fields 1–16): meaning and data type. Source: Antonio et al. (2019); code: data_dictionary.ipynb.

Field	Meaning	Data type (course taxonomy)	Order	Temporal or geographic
is_repeated_guest	The booking name belongs to a repeated guest (1) or not (0)	Categorical, nominal (binary)	No order	Neither
previous_cancellations	Earlier bookings of the same customer that were cancelled	Numerical, discrete, ratio	Sequential	Neither
previous_bookings_not_canceled	Earlier bookings of the same customer that were not cancelled	Numerical, discrete, ratio	Sequential	Neither
reserved_room_type	Code of the room type reserved (letters, anonymised)	Categorical, nominal	No order	Neither
assigned_room_type	Code of the room type assigned; can differ from the reserved one (overbooking, guest request)	Categorical, nominal	No order	Neither
booking_changes	Changes made to the booking until check-in or cancellation	Numerical, discrete, ratio	Sequential	Neither
deposit_type	Deposit to guarantee the booking: No Deposit, Non Refund (covers the whole stay) or Refundable (covers part of it)	Categorical, nominal	No order	Neither
agent	ID of the travel agency that made the booking; NULL = not made by an agency	Categorical, nominal (identifier)	No order	Neither
company	ID of the company that made or pays the booking; NULL = no company	Categorical, nominal (identifier)	No order	Neither
days_in_waiting_list	Days the booking waited before it was confirmed to the customer	Numerical, discrete, ratio	Sequential	Temporal (duration)
customer_type	Type of booking: Transient, Transient-Party, Contract or Group	Categorical, nominal	No order	Neither
adr	Average daily rate: lodging revenue of the booking divided by its number of nights	Numerical, continuous, ratio	Sequential	Neither
required_car_parking_spaces	Car parking spaces the customer asked for	Numerical, discrete, ratio	Sequential	Neither
total_of_special_requests	Special requests made by the customer (for example a twin bed or a high floor)	Numerical, discrete, ratio	Sequential	Neither
reservation_status	Last status of the booking: Canceled, Check-Out or No-Show. Repeats is_canceled	Categorical, nominal	No order	Neither
reservation_status_date	Date on which the last status was set	Numerical, discrete, interval (date)	Sequential	Temporal

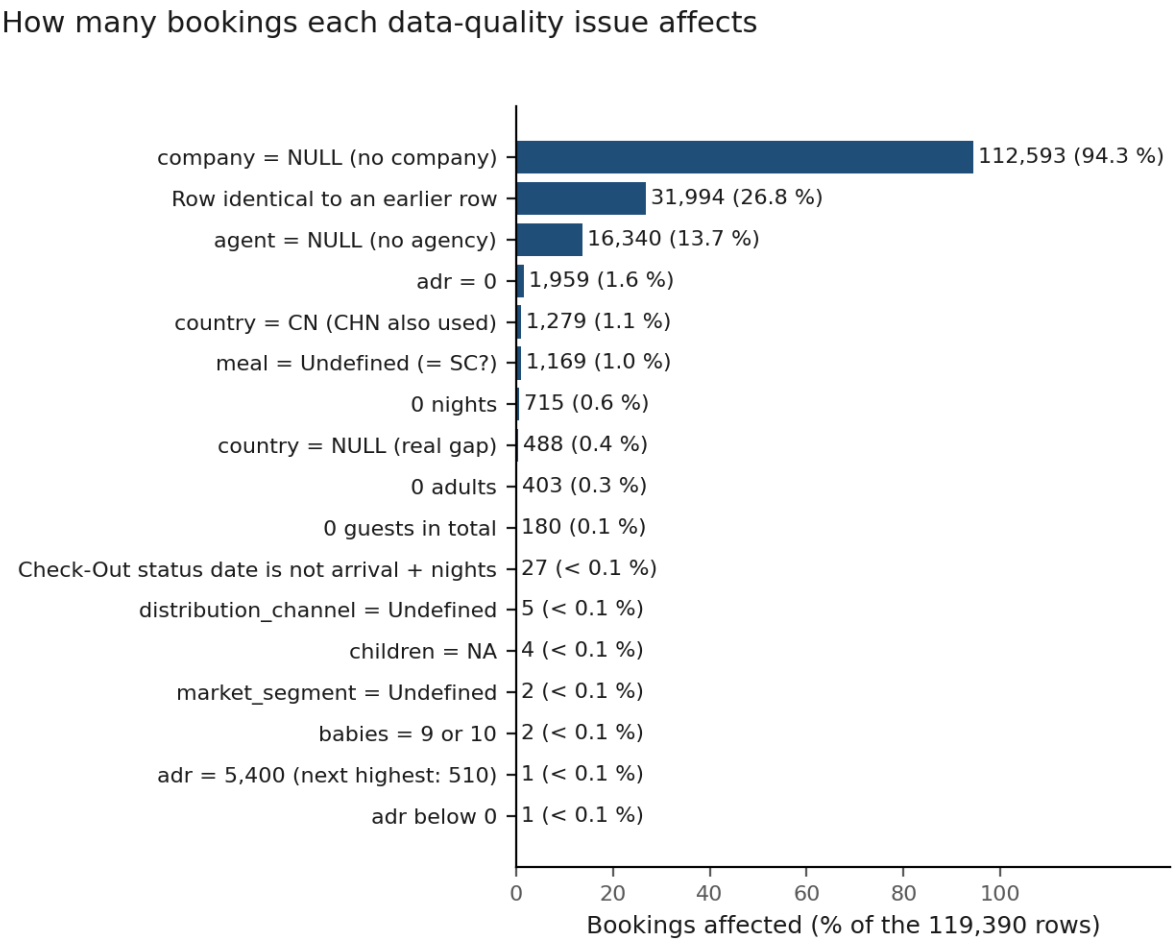
Figure 2. Data dictionary A (fields 17–32): meaning and data type. Source: as Figure 1.

Field	Range or domain (raw data)	Units	Precision needed	# Missing	Missing placeholder
hotel	2: City Hotel, Resort Hotel	No unit	Label	0	—
is_canceled	0 = no, 1 = yes	No unit	Count; rate in % with 1 decimal	0	—
lead_time	0 to 737	Days	Whole days	0	—
arrival_date_year	2015 to 2017	Year	Year	0	—
arrival_date_month	January to December (12 months)	Month	Month	0	—
arrival_date_week_number	1 to 53	Week of the year	Week	0	—
arrival_date_day_of_month	1 to 31	Day of the month	Day	0	—
stays_in_weekend_nights	0 to 19	Nights	Whole nights	0	—
stays_in_week_nights	0 to 50	Nights	Whole nights	0	—
adults	0 to 55	Persons	Whole persons	0	—
children	0 to 10	Persons	Whole persons	4	NA
babies	0 to 10	Persons	Whole persons	0	—
meal	5: BB, HB, SC, Undefined, FB	No unit	Label	1,169	Undefined
country	177 codes (most frequent: PRT, GBR, FRA)	No unit	Country	488	NULL
market_segment	8: Online TA, Offline TA/TO, Groups, Direct, Corporate, Complementary, Aviation, Undefined	No unit	Label	2	Undefined
distribution_channel	5: TA/TO, Direct, Corporate, GDS, Undefined	No unit	Label	5	Undefined

Figure 3. Data dictionary B (fields 1–16): values, units, precision and missing values. Source: as Figure 1.

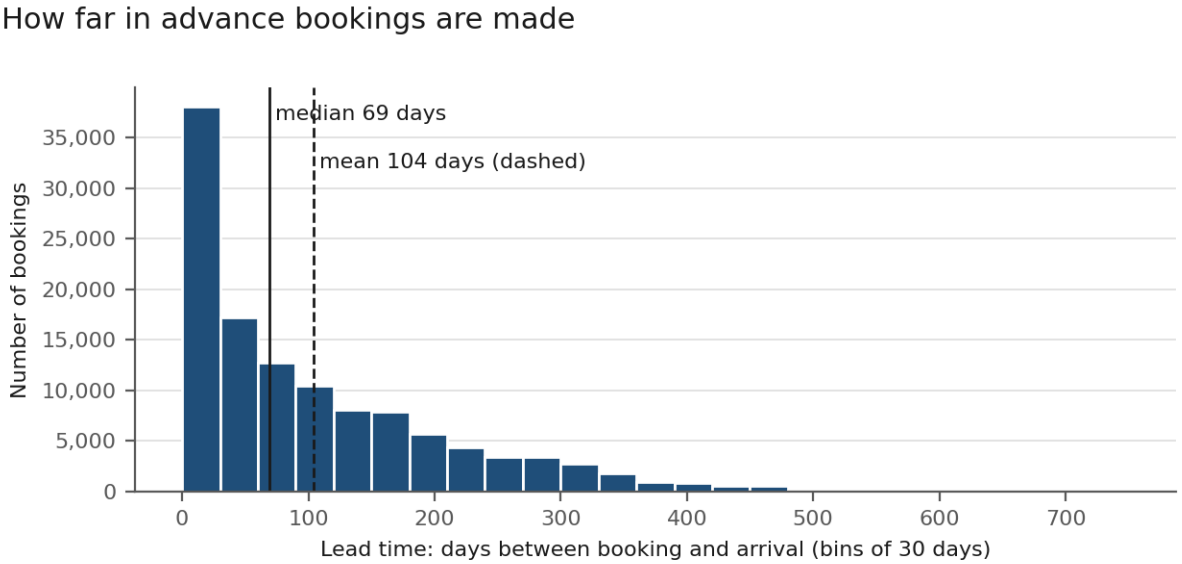
Field	Range or domain (raw data)	Units	Precision needed	# Missing	Missing placeholder
is_repeated_guest	0 = no, 1 = yes	No unit	Count; share in % with 1 decimal	0	—
previous_cancellations	0 to 26	Bookings	Whole bookings	0	—
previous_bookings_not_canceled	0 to 72	Bookings	Whole bookings	0	—
reserved_room_type	10: A, D, E, F, G, B, C, H, P, L	No unit	Label	0	—
assigned_room_type	12: A, D, E, F, G, C, B, H, I, K, P, L	No unit	Label	0	—
booking_changes	0 to 21	Changes	Whole changes	0	—
deposit_type	3: No Deposit, Non Refund, Refundable	No unit	Label	0	—
agent	333 IDs, from 1 to 535	No unit	Label	16,340	NULL
company	352 IDs, from 6 to 543	No unit	Label	112,593	NULL
days_in_waiting_list	0 to 391	Days	Whole days	0	—
customer_type	4: Transient, Transient-Party, Contract, Group	No unit	Label	0	—
adr	-6.38 to 5400	Money per night (currency not stated; probably euros)	Whole euros (file has 2 decimals)	0	—
required_car_parking_spaces	0 to 8	Parking spaces	Whole spaces	0	—
total_of_special_requests	0 to 5	Requests	Whole requests	0	—
reservation_status	3: Check-Out, Canceled, No-Show	No unit	Label	0	—
reservation_status_date	2014-10-17 to 2017-09-14	Date (YYYY-MM-DD)	Day	0	—

Figure 4. Data dictionary B (fields 17–32). Source: as Figure 1. Accessible version: data_dictionary.csv.



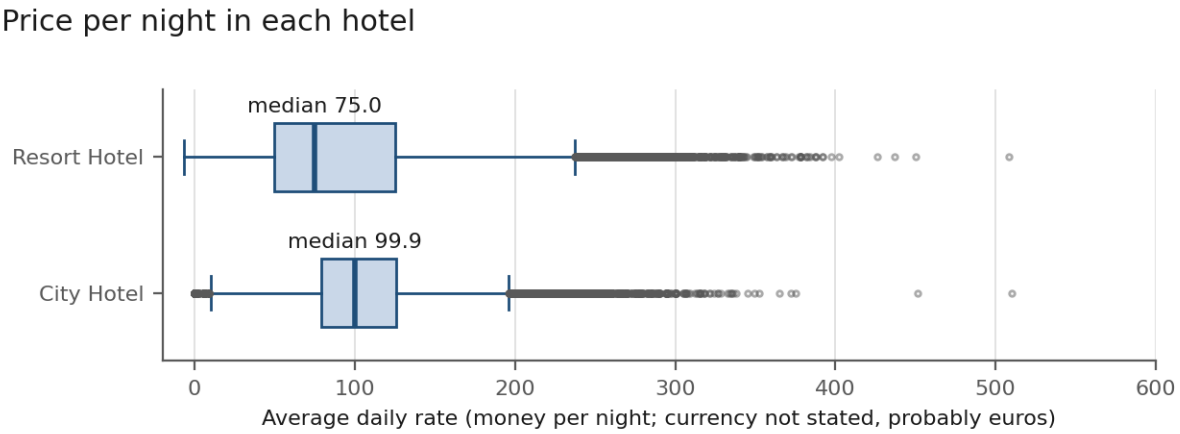
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file.

Figure 5. Bookings affected by each data-quality issue. Source: raw data; code: data_quality.ipynb.



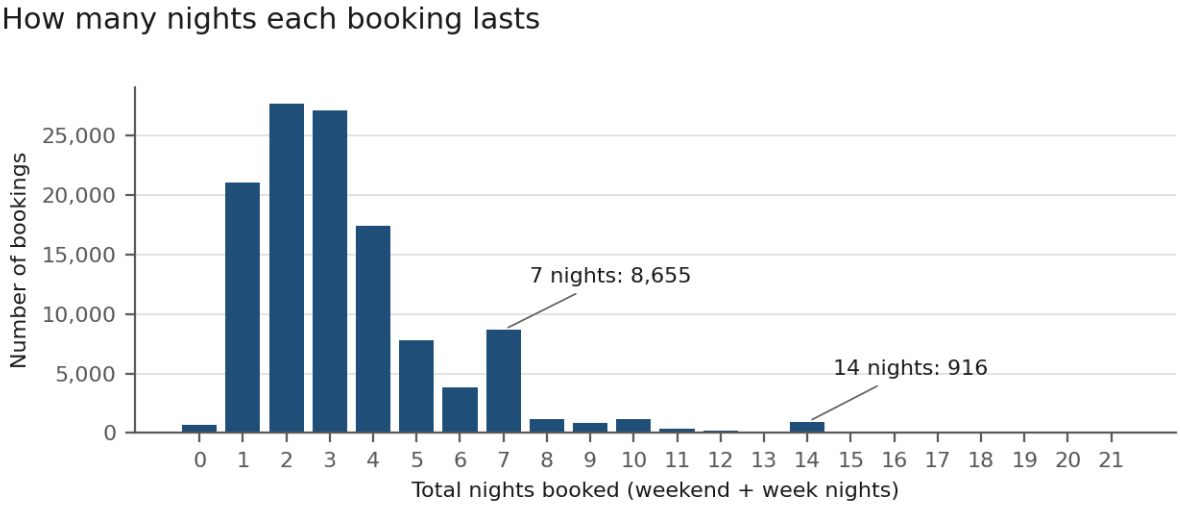
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 6. Lead time distribution with median and mean. Source: raw data; code: distributions_outliers.ipynb.



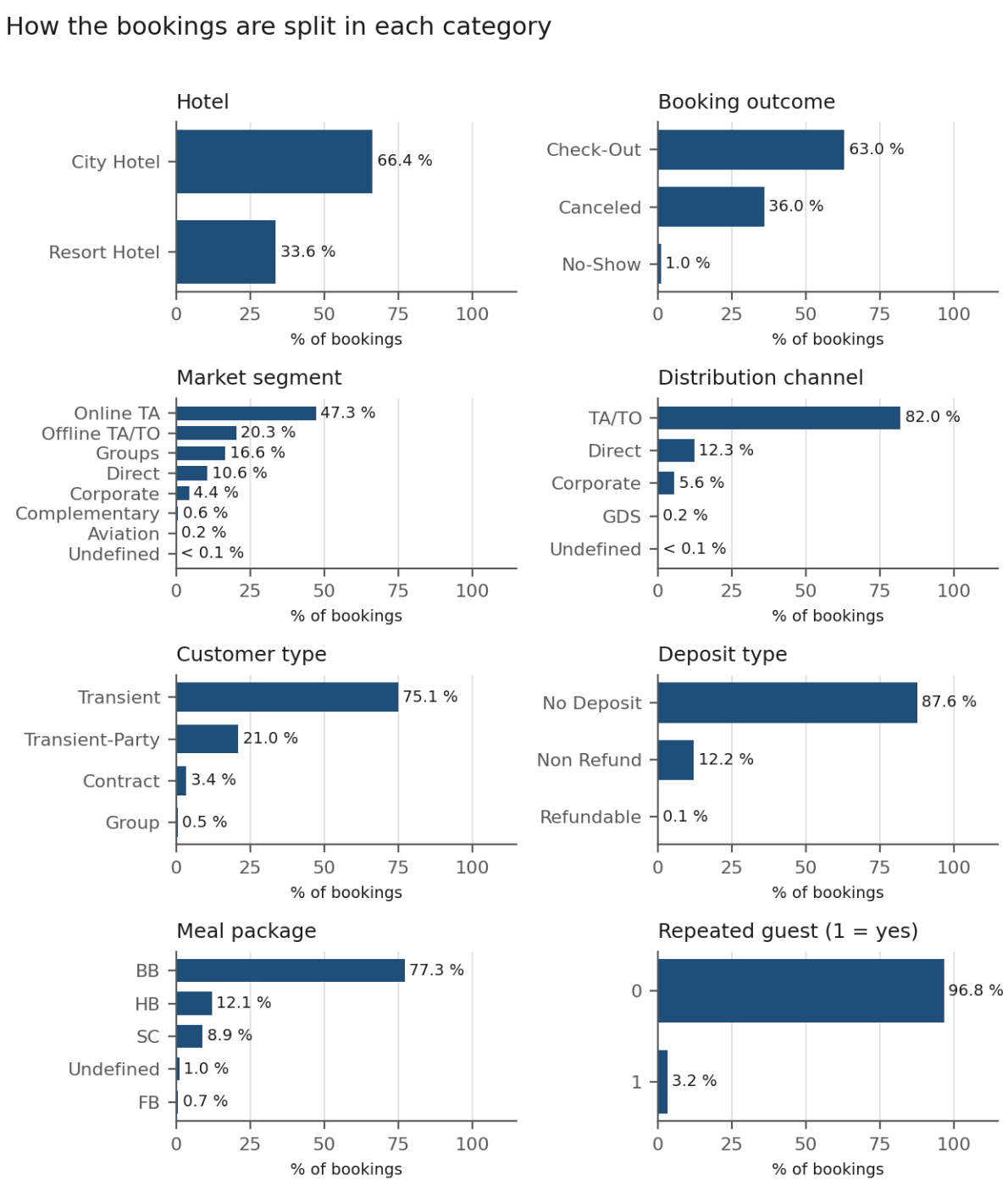
Not shown: 1 booking with adr = 5,400. Boxes: middle half of the bookings; whiskers: 1.5 × IQR; circles: values beyond.
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 7. Price per night by hotel. Source: as Figure 6.



Not shown: 162 bookings of more than 21 nights (maximum 69 nights).
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

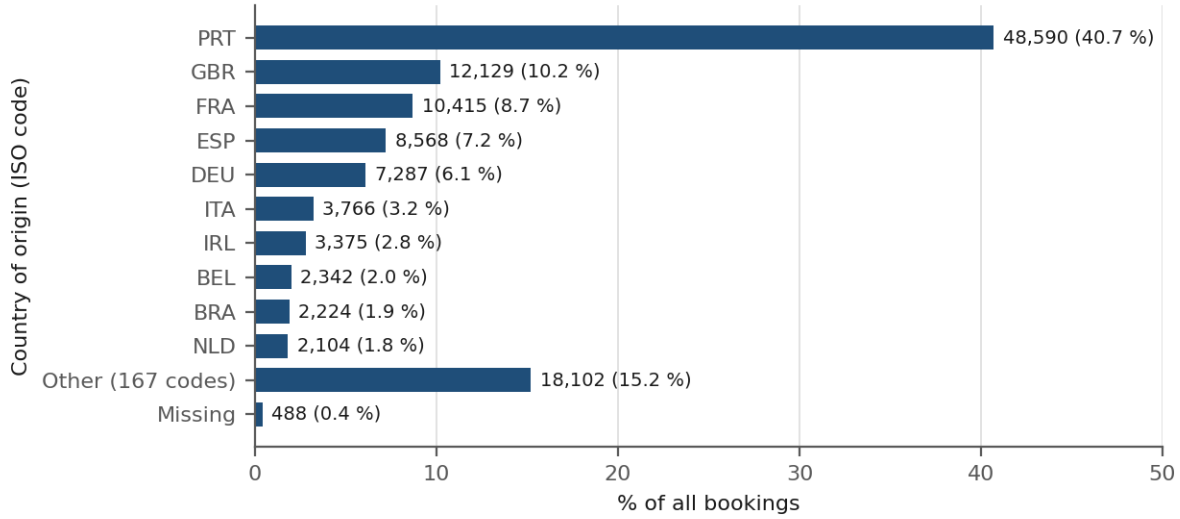
Figure 8. Nights per booking. Source: as Figure 6.



Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 9. Shares of the categorical fields. Source: as Figure 6.

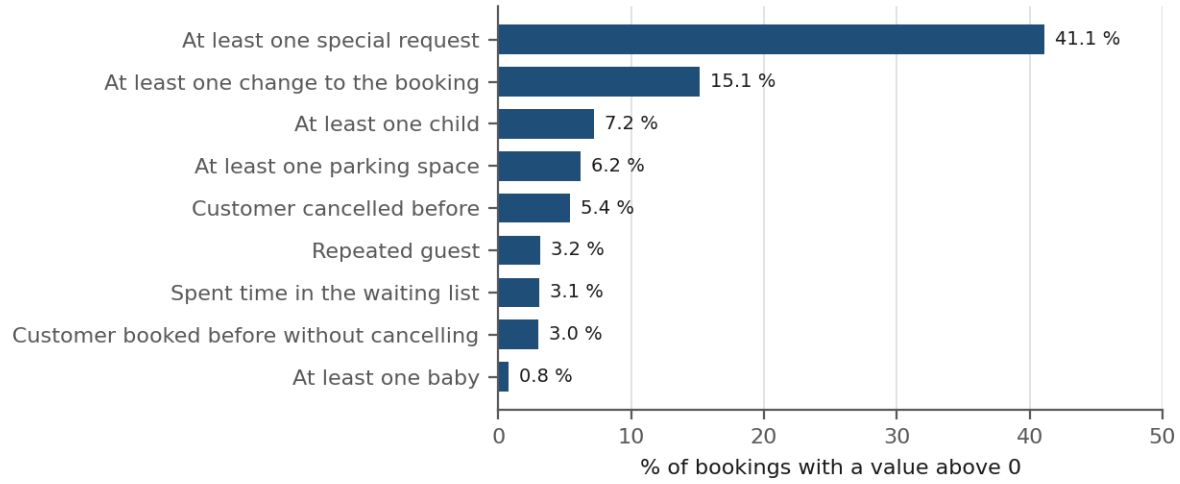
Where the guests come from: ten main countries



Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

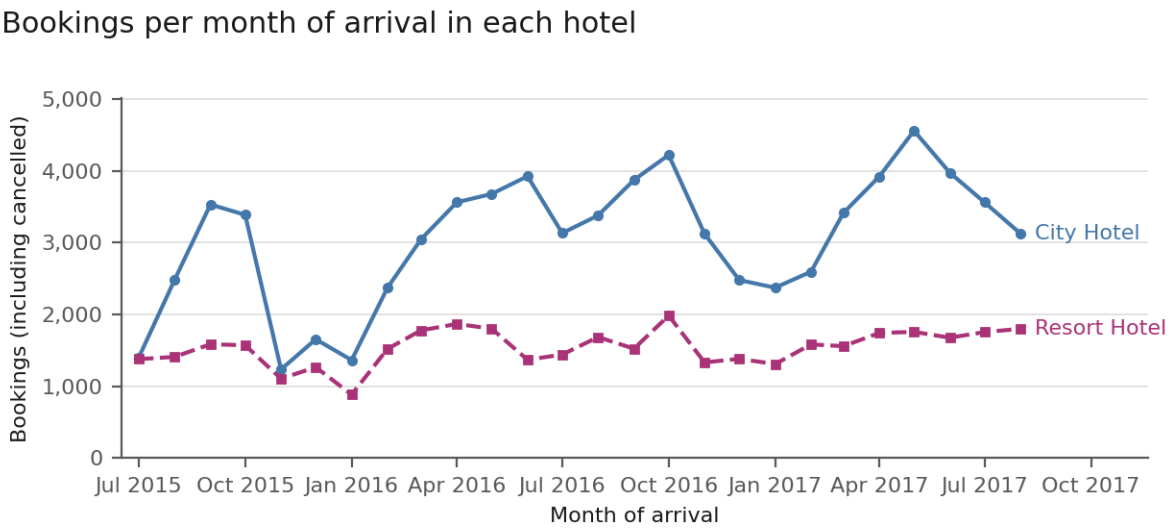
Figure 10. Country of origin, ten main countries. Source: as Figure 6.

Most booking details are rare



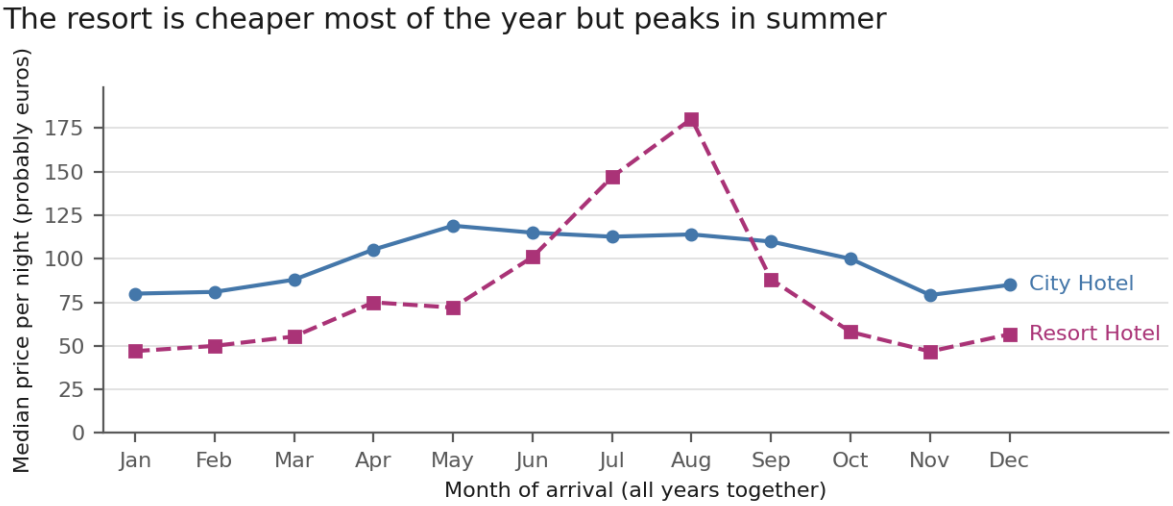
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 11. Share of bookings with each detail. Source: as Figure 6.



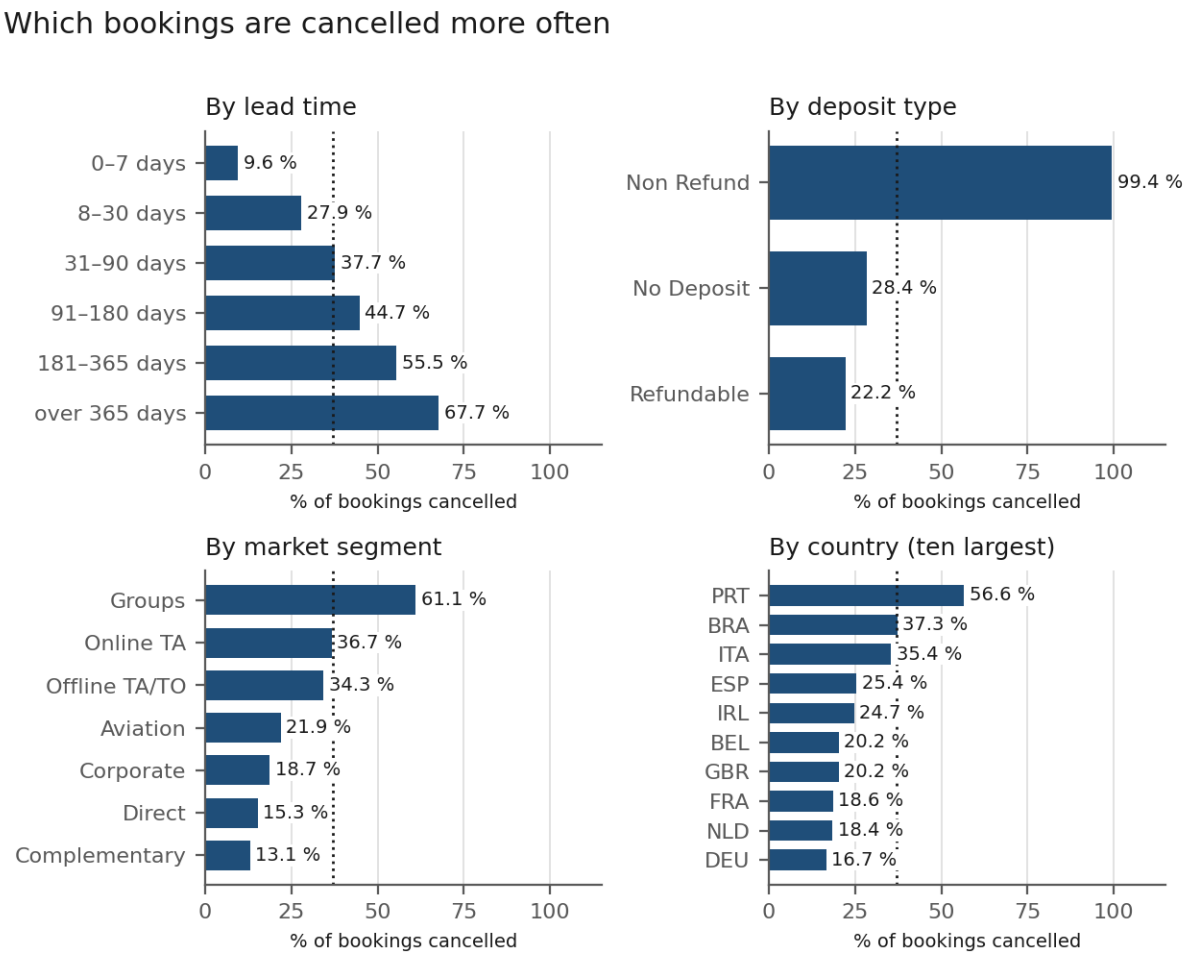
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 12. Bookings per month of arrival by hotel. Source: raw data; code: relationships.ipynb.



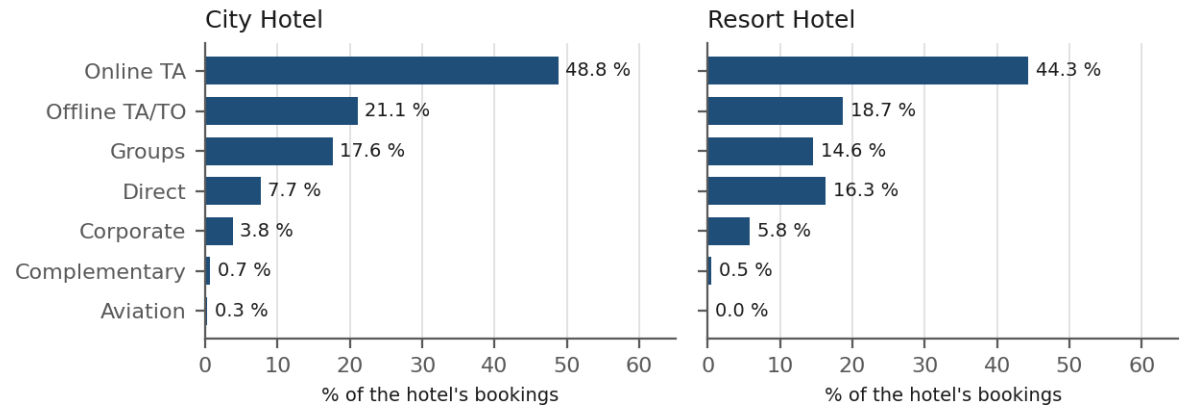
Only stays that were not cancelled and have a price above 0.
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 13. Median price per night by month and hotel. Source: as Figure 12.



Dotted line: rate over all bookings (37.0 %). Raw data, duplicates included.
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

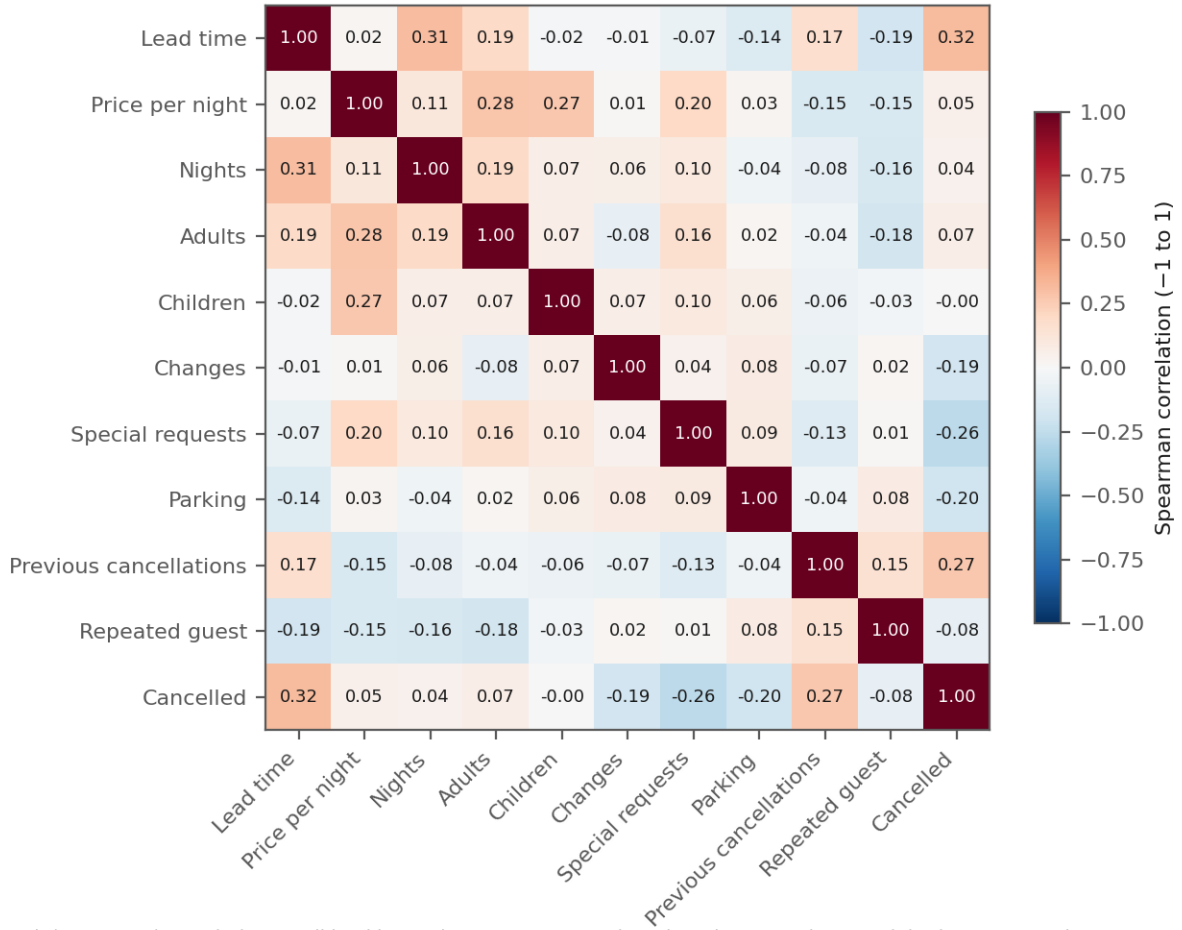
Figure 14. Cancellation rate by lead time, deposit, segment and country. Source: as Figure 12.



Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 15. Market segments in each hotel. Source: as Figure 12.

How the numeric fields move together



Rank (Spearman) correlation on all bookings. Blue: one goes up when the other goes down; red: both go up together.
Source: Antonio, de Almeida and Nunes (2019), hotel bookings dataset, raw file (119,390 bookings).

Figure 16. Spearman correlation between numeric fields. Source: as Figure 12.

Reference

Antonio, N., de Almeida, A. and Nunes, L. (2019). Hotel booking demand datasets. *Data in Brief*, 22, 41–49. https://doi.org/10.1016/j.dib.2018.11.126